MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

OPTICAL ISOLATOR, DIGITAL, KEY CODE MUM

This specification is approved for use within the Space and Naval Warfare Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic optical isolator module.

1.2 Classification. This specification covers one class of modules.

Class I as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standards. The following specification and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of the specification and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets, or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Space and Naval Warfare Systems Command, ATTN: SPAWAR 8111, Department of the Navy, Washington, DC 20363, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5963

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 202640-20T17:01Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 1.672 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be MUM.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/246-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 5.0 ±0.5 V dc.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

Source: https://assist.dla.mil -- Downloaded: 202640-20T17:01Z
Check the source to verify that this is the current version before use.

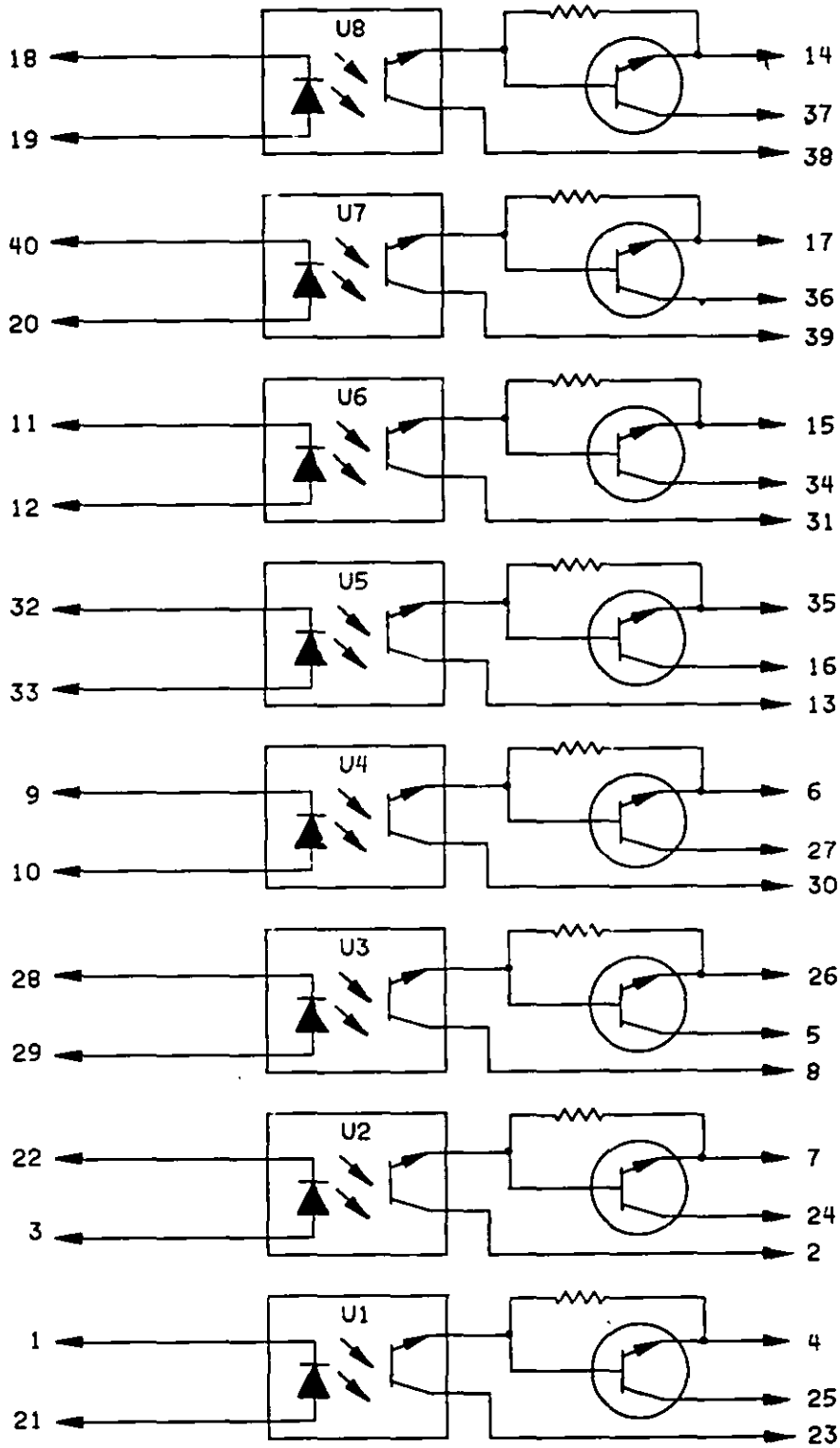


FIGURE 1. MUM module functional diagram, optical isolator.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:01Z
Check the source to verify that this is the current version before use.

* TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°C ±5°C ambient temperature				Operating temperature 0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1-A16	Maximum stress voltage	II	S	---	---	---	---	---	---	---	---	
B1	Isolation	III	S	10	∞	1	∞	10	∞	1	∞	mΩ
B2-B9	Isolation (circuit)	III	X	10	∞	1	∞	10	∞	1	∞	mΩ
C1-C8	Common mode voltage (measuring output leakage current)	IV	Q	0	50	0	50	0	50	0	50	nA
D1-D8	LED forward voltage	V	S	0.4	1.5	0.4	1.5	0.4	1.5	0.4	1.5	V
E1-E8	Output leakage current	VI	X	0	40	0	40	0	40	0	40	µA
E9-E16	Input threshold current	VI	X	0.1	2.2	0.1	2.2	0.1	2.2	0.1	2.2	mA
F1-F8	LED reverse current	VII	S	0	150	0	150	0	600	0	600	µA
G1-G8	Output voltage, low level, I_{OL} = 80 mA, If = 16 mA	VIII	X	0	400	0	400	0	400	0	400	mV
G9-G16	Output voltage, low level, I_{OL} = 20 mA, If = 4 mA	VIII	S	0	400	0	400	0	400	0	400	mV
G17-G24	Output voltage, low level, I_{OL} = 8 mA, If = 2.6 mA	VIII	S	0	400	0	400	0	400	0	400	mV
H1-H8	Output saturation voltage (I_{OL} = 125 mA)	IX	X	0	400	0	400	0	400	0	400	mV
J1	Power supply current, LED off	X	S	0	800	0	800	0	800	0	800	µA
J2	Power supply current, LED on	X	S	0	200	0	200	0	200	0	200	mA
K1-K8	Transition time (t_{TLH})	XI	S	1	12	1	12	1	17	1	17	µs
L1-L8	Transition time (t_{THL})	XI	S	0.060	1.400	0.060	1.400	0.065	1.800	0.065	1.800	µs
M1-M8	Propagation delay time (t_{PLH})	XI	S	1	14	1	14	1	18	1	18	µs

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:01Z
Check the source to verify that this is the current version before use.

* TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°C ±5°C ambient temperature				Operating temperature 0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
N1-N8	Propagation delay time (t_{PHL})	XI	S	0.1	2.5	0.1	2.5	0.1	3.5	0.1	3.5	µs
P1-P8	Output crosstalk, low level	XII	Q	0	400	0	400	0	400	0	400	mV
Q1-Q8	Output crosstalk, high level	XIII	Q	0	400	0	400	0	400	0	400	mV
R1-R8	Input capacitance	XIV	Q	1	200	1	200	1	200	1	200	pF

1/ "X" parameters shall be 100 percent tested during module acceptance tests.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from the inspection lot, whichever is higher) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:01Z
Check the source to verify that this is the current version before use.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

* 4.4.2 Order of testing. Test conditions A through R shall be performed in any order with the following exceptions. Test conditions A and B are stress tests and will be performed only at 25°C ±5°C. If test conditions A or B are part of a sequence of tests, the order of testing shall be as follows: B, A where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through R in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions C through Q shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column, shall be performed using the initial limits at 25°C ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests C through Q shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests C through R shall be performed in the order specified in 4.4.2 using the 25°C ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Cross reference of classification. Part number M28787/246-1 supersedes part number M28787/246. The two parts are interchangeable.

* 6.4 Module description. The following electrical parameters reflect the 0°C to 60°C end-of-life limits from table I.

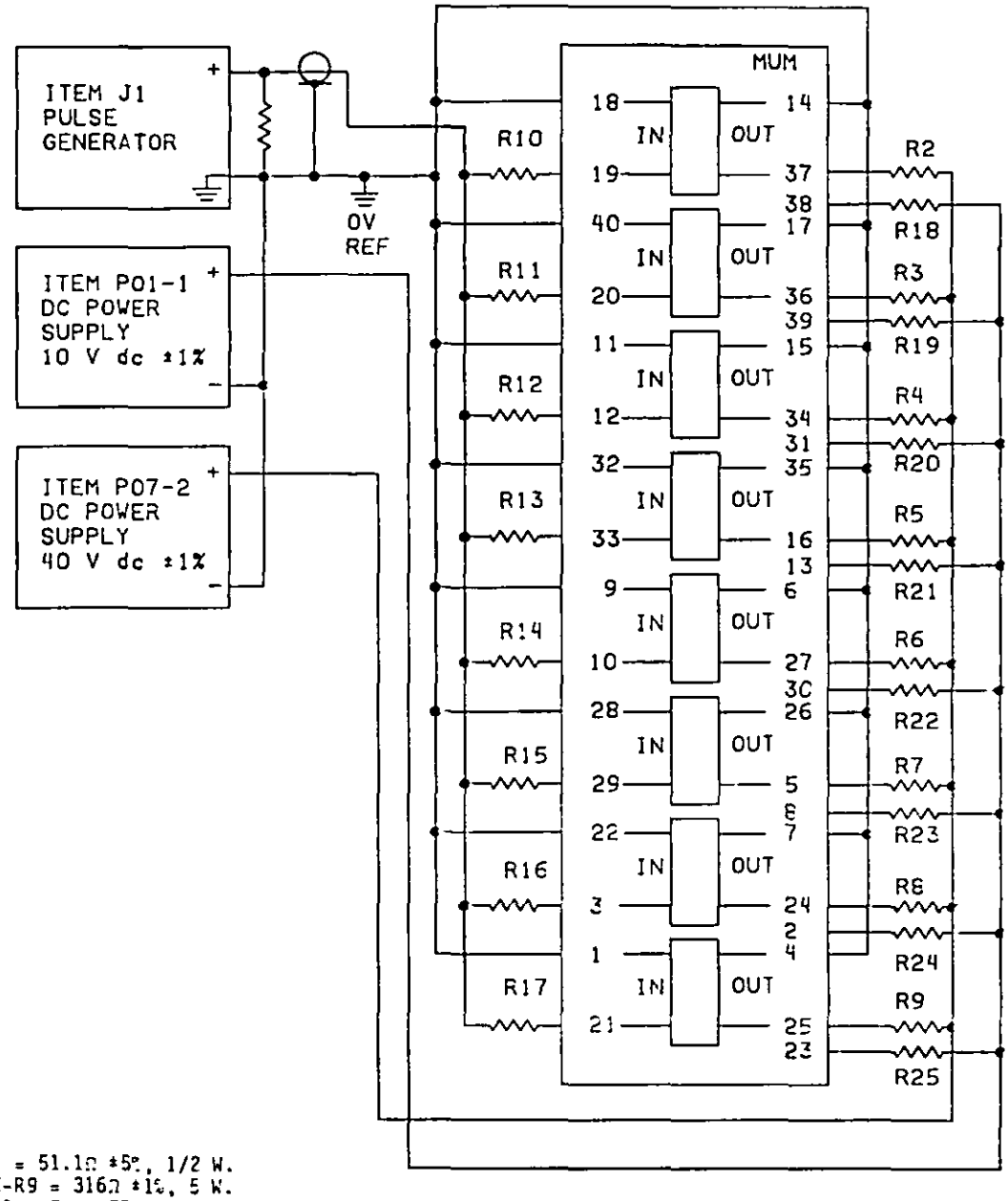
Maximum power dissipation	1.5 W (outputs not loaded)
Number of complete circuits per module	8

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:01Z
Check the source to verify that this is the current version before use.

I_{CC} (V_{CC} = 5.5 V)	200 mA maximum
Input reverse current ($V_{Reverse}$ = 3 V dc)	600 µA maximum
Input forward voltage ($I_{Forward}$ = 26 mA)	1.5 V dc maximum
Input reverse voltage	3 V dc maximum
I_{IH} ($V_{Forward}$ = 1.5 V dc) (absolute maximum continuous current)	26 mA maximum
Input threshold current (I_f = 2 mA)	100 µA minimum
Output saturation voltage (V_{CE}) (I_C = 125 mA)	400 mV dc maximum
Low-level output drive current (V_{IN} = 5 V dc, $I_{Forward}$ = 26 mA, V_{OL} = 400 mV dc)	125 mA maximum
Low-level output drive current (V_{IN} = 5 V dc, $I_{Forward}$ = 16 mA, V_{OL} = 400 mV dc)	80 mA maximum
Low-level output drive current (V_{IN} = 5 V dc, $I_{Forward}$ = 4 mA, V_{OL} = 400 mV dc)	20 mA maximum
Low-level output drive current (V_{IN} = 5 V dc, $I_{Forward}$ = 2.6 mA, V_{OL} = 400 mV dc)	8 mA maximum
Common mode voltage (each input to any other point)	100 V dc
Propagation delay time (t_{PLH})	18 µs maximum
Propagation delay time (t_{PHL})	3.5 µs maximum
Transition time (t_{TLH})	17 µs maximum
Transition time (t_{THL})	1.8 µs maximum
Opto-isolator breakdown voltage (B_{CEO})	30 V minimum
Transistor breakdown voltage (B_{CEO})	40 V minimum
Opto-isolator power dissipation	162 mW maximum
Transistor power dissipation	475 mW maximum

6.5 Module function. This module contains eight optical isolators, each having a light emitter diode (LED) optically coupled to a photosensitive amplifier. The input network for each circuit is independent of all others with direct access to the LED for operation over a range of input voltages. The output for each circuit is a collector-open transistor to permit operation over a wide range of voltages. The module is TTL compatible when connected as shown on figure 5.

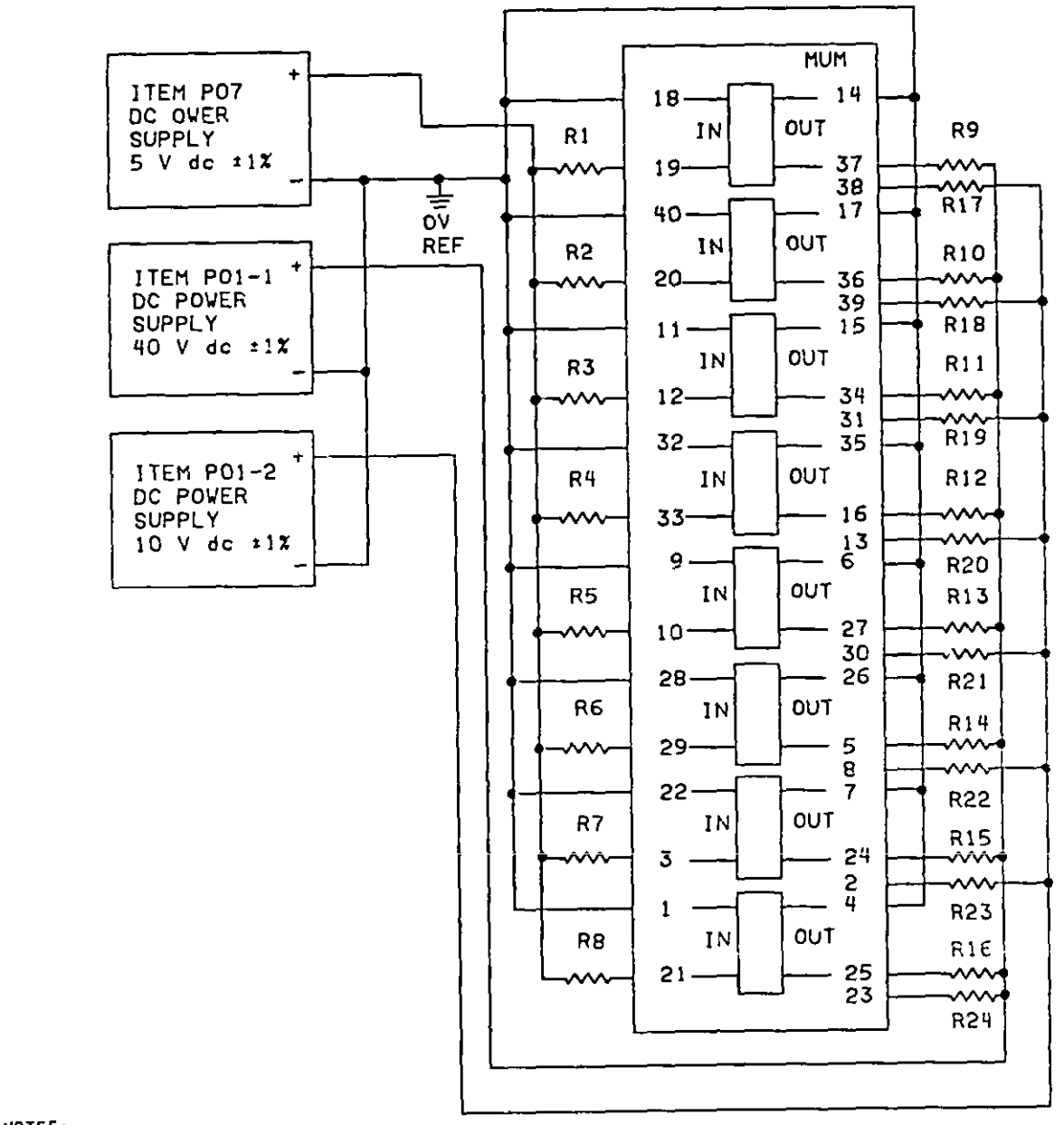
Source: https://assist.dla.mil -- Downloaded: 202610-20T17:01Z
Check the source to verify that this is the current version before use.



NOTES:
1. R1 = 51.1Ω ±5%, 1/2 W.
2. R2-R9 = 316Ω ±1%, 5 W.
3. R10-R17 = 177Ω ±1%, 1/4 W.
4. R18-R25 = 2.15 kΩ ±1%, 1/4 W.
5. Item J01 input pulse parameters:
   PW = 50 ±10 µs.
   PA = 5.0 ±0.5 V dc.
   PRP = 100 ±20 µs.
   $t_{TLH} = t_{THL}$ = 25 ±15 ns.

FIGURE 2. Operating temperature and life test circuit.

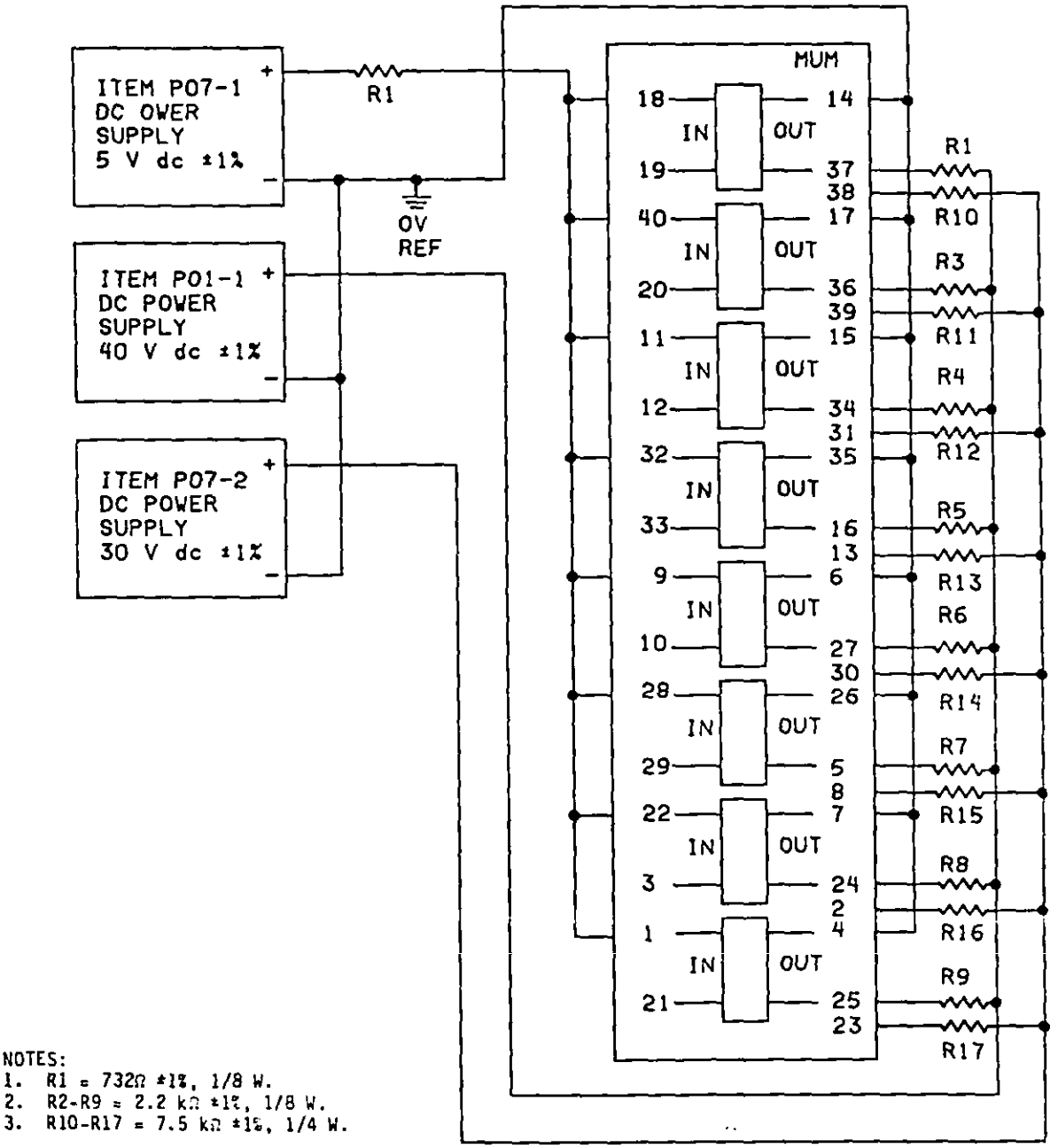
Source: https://assist.dla.mil -- Downloaded: 202610-20T17:01Z
Check the source to verify that this is the current version before use.



NOTES:
1. R1-R8 = 177Ω ±1%, 1/4W.
2. R9-R16 = 316Ω ±1%, 5 W.
3. R17-R24 = 613Ω ±1%, 1/4 W.

FIGURE 3. Component temperature test circuit.

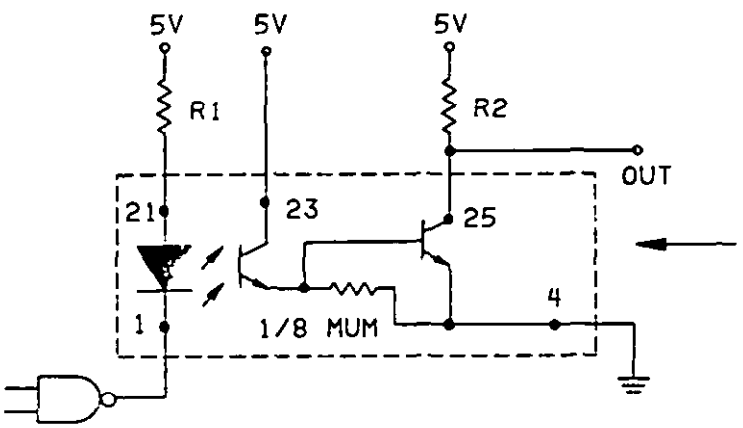
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:01Z
Check the source to verify that this is the current version before use.



NOTES:
1. R1 = 732Ω ±1%, 1/8 W.
2. R2-R9 = 2.2 kΩ ±1%, 1/8 W.
3. R10-R17 = 7.5 kΩ ±1%, 1/4 W.

FIGURE 4. Vibration test circuit.

Source: https://assist.dla.mil -- Downloaded: 202610-20T17:01Z
Check the source to verify that this is the current version before use.



NOTES:
1. R1 = 191 Ω.
2. R2 = 2.2 kΩ.

FIGURE 5. Circuit for TTL compatibility.

Source: https://assist.dla.mil -- Downloaded: 20260-20T17:01Z
Check the source to verify that this is the current version before use.

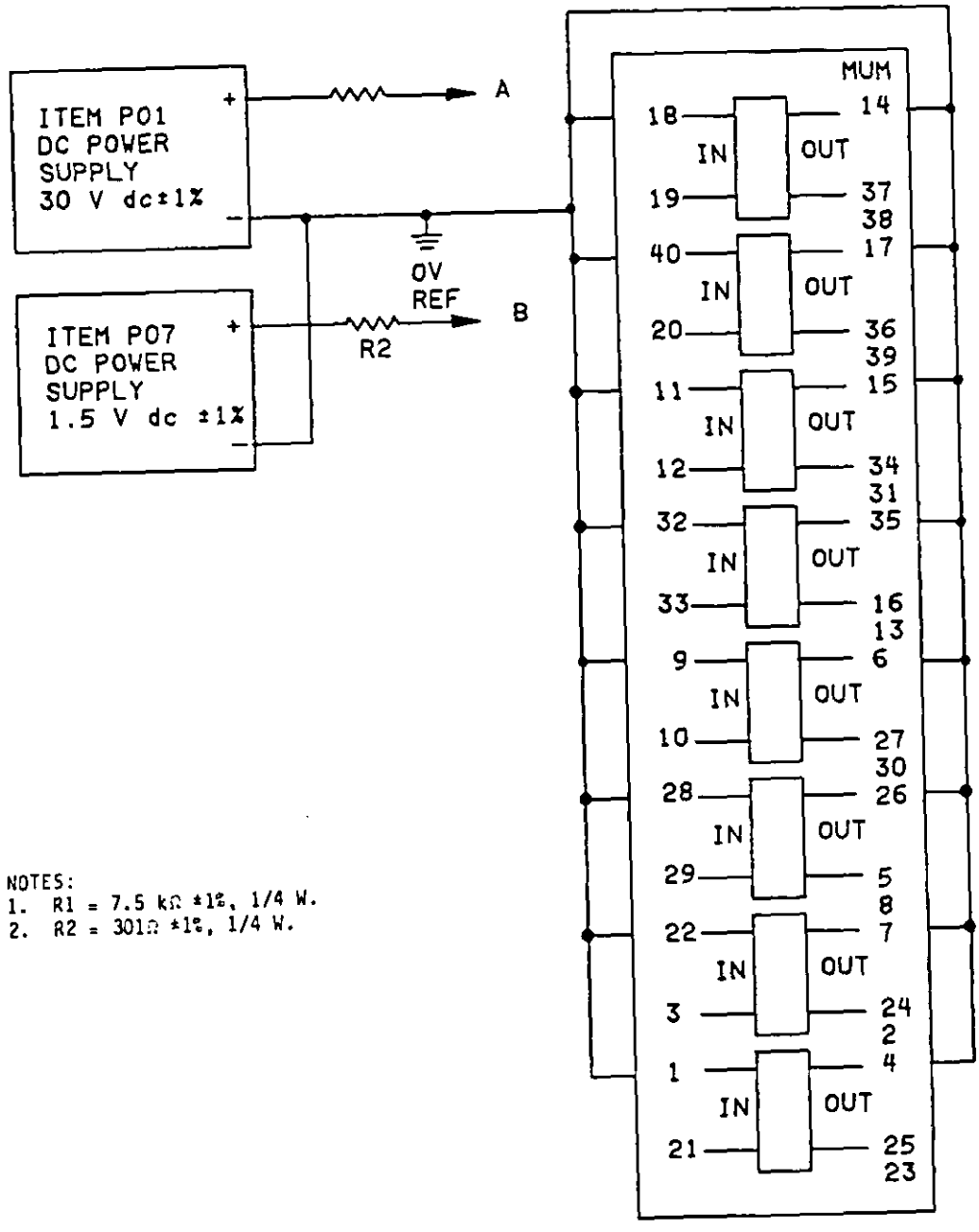
TABLE II. Maximum stress voltage test. 1/, 2/

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A1	Maximum stress voltage	23	Open	2/	2/	2/	6
A2	"	2	"	"	"	"	"
A3	"	8	"	"	"	"	"
A4	"	30	"	"	"	"	"
A5	"	13	"	"	"	"	"
A6	"	31	"	"	"	"	"
A7	"	39	"	"	"	"	"
A8	"	38	"	"	"	"	"
A9	"	23	21	"	"	"	"
A10	"	2	3	"	"	"	"
A11	"	8	29	"	"	"	"
A12	"	30	10	"	"	"	"
A13	"	13	33	"	"	"	"
A14	"	31	12	"	"	"	"
A15	"	39	20	"	"	"	"
A16	"	38	19	"	"	"	"

1/ For test conditions A1 through A16 make all circuit connections as specified herein and on figure 2 for a minimum of 1 minute. No electrical measurement is to be taken.

2/ Module must pass all remaining tests after this test is performed.

Source: https://assist.dla.mil -- Downloaded: 2026-02-20T17:01Z
Check the source to verify that this is the current version before use.



NOTES:
1. R1 = 7.5 kΩ ±1%, 1/4 W.
2. R2 = 301Ω ±1%, 1/4 W.

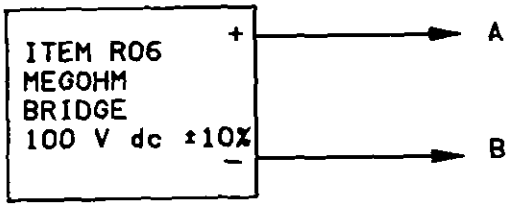
FIGURE 6. Maximum stress voltage test circuit for table II.

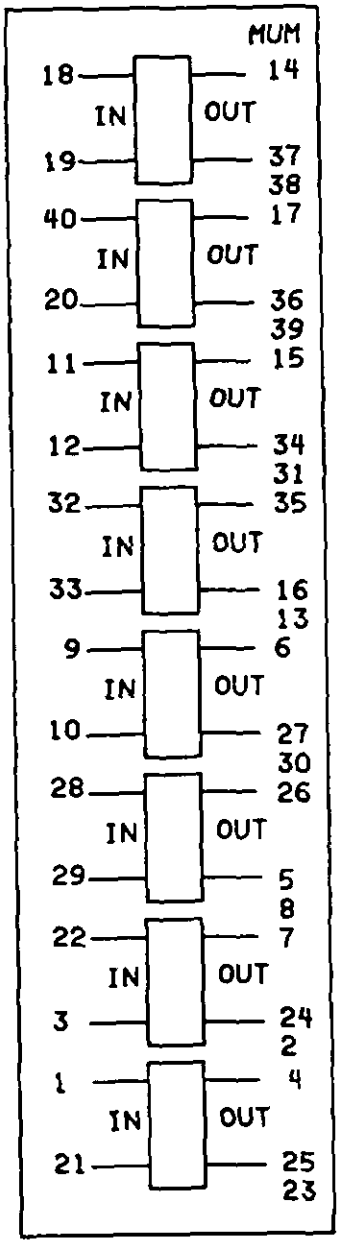
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:01Z
Check the source to verify that this is the current version before use.

TABLE III. Isolation test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
B1	Isolation	1 thru 40	Frame	10	∞	MΩ	7
B2	Isolation (circuit)	1, 21	2 thru 20 22 thru 40	"	"	"	"
B3	"	3, 22	1, 2, 4, thru 21 23 thru 40	"	"	"	"
B4	"	28, 29	1 thru 27 30 thru 40	"	"	"	"
B5	"	9, 10	1 thru 8 11 thru 40	"	"	"	"
B6	"	32, 33	1 thru 31 34 thru 40	"	"	"	"
B7	"	11, 12	1 thru 10 13 thru 40	"	"	"	"
B8	"	20, 40	1 thru 19 21 thru 39	"	"	"	"
B9	"	18, 19	1 thru 17 20 thru 40	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:01Z
Check the source to verify that this is the current version before use.





WARNING:

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the function switch of the megohm bridge at "discharge" before connecting or disconnecting the test leads.

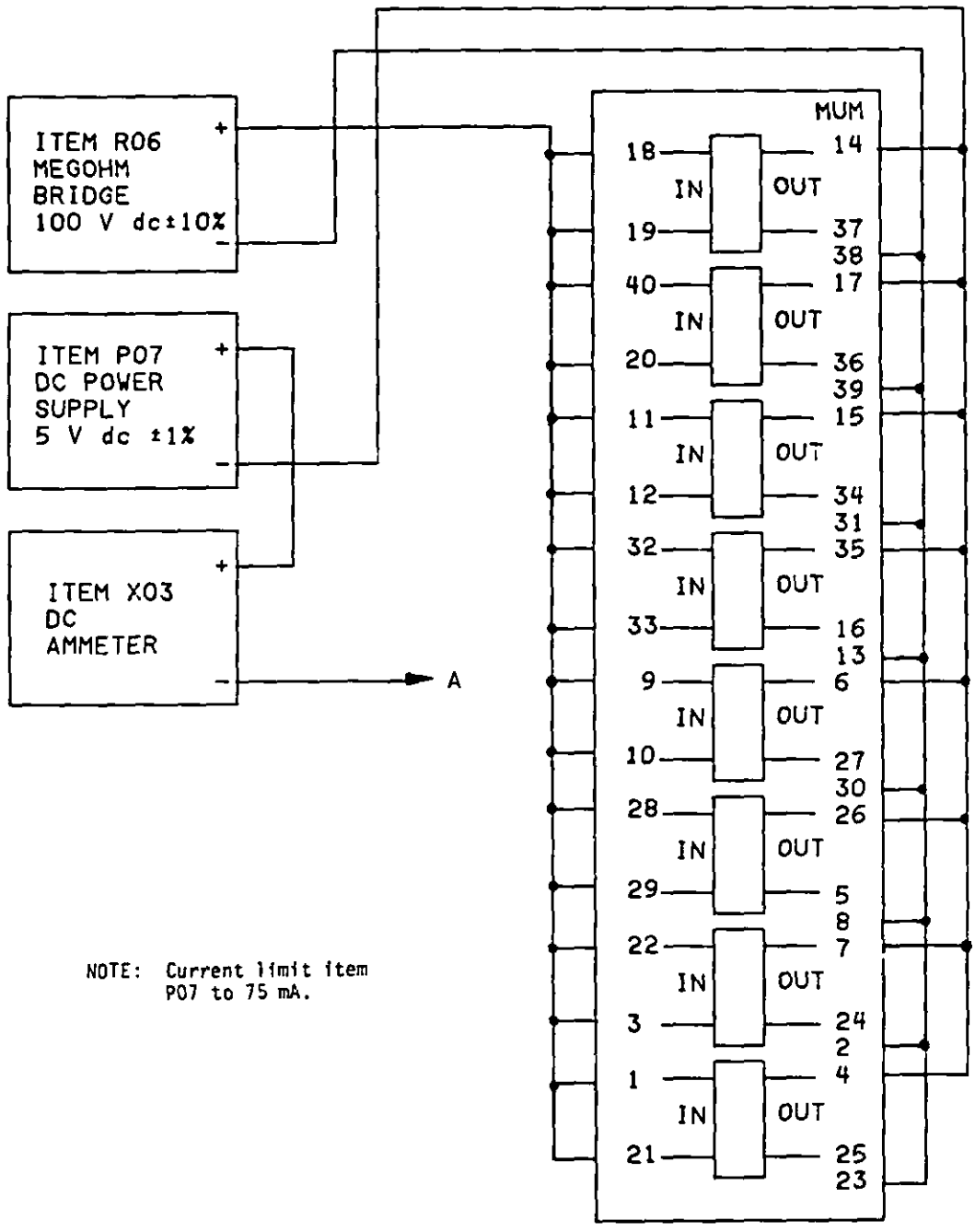
FIGURE 7. Isolation test circuit for table III.

Source: https://assist.dla.mil -- Downloaded: 202610-20T17:01Z
Check the source to verify that this is the current version before use.

TABLE IV. Common mode voltage (measuring output leakage current) test.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
C1	Common mode voltage (measuring output leakage current)	37	0	50	nA	8
C2	"	36	"	"	"	"
C3	"	34	"	"	"	"
C4	"	16	"	"	"	"
C5	"	27	"	"	"	"
C6	"	5	"	"	"	"
C7	"	24	"	"	"	"
C8	"	25	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-02-20T17:01Z
Check the source to verify that this is the current version before use.



NOTE: Current limit item P07 to 75 mA.

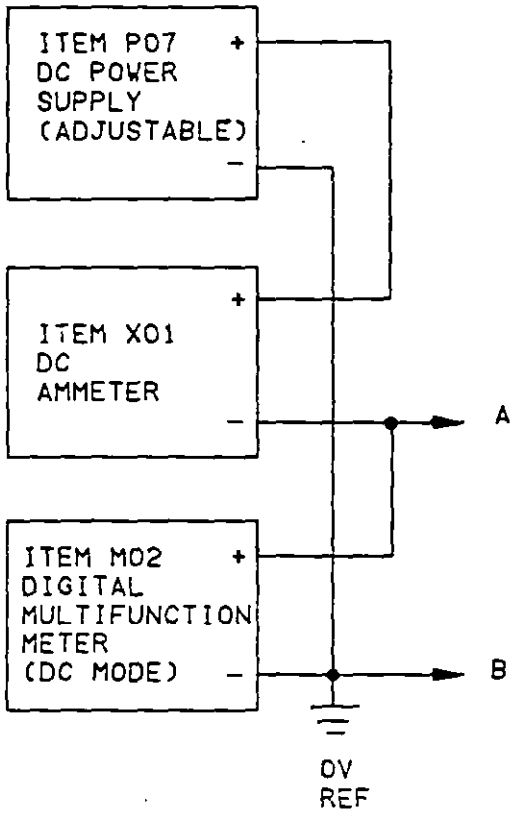
FIGURE 8. Common mode voltage (measuring output leakage current) test circuit for table IV.

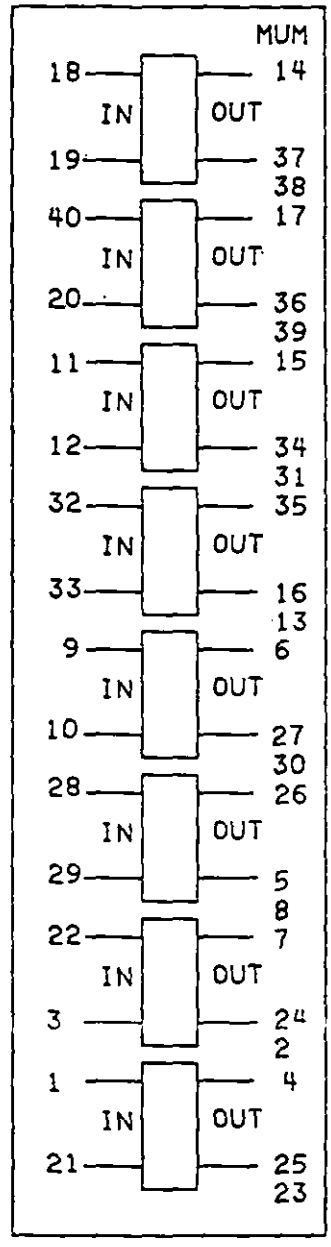
Source: https://assist.dla.mil -- Downloaded: 202610-20T17:01Z
Check the source to verify that this is the current version before use.

TABLE V. LED forward voltage test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
D1	LED forward voltage	19	18	0.4	1.5	V	9
D2	"	20	40	"	"	"	"
D3	"	12	11	"	"	"	"
D4	"	33	32	"	"	"	"
D5	"	10	9	"	"	"	"
D6	"	29	28	"	"	"	"
D7	"	3	22	"	"	"	"
D8	"	21	1	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 20260-20T17:01Z
Check the source to verify that this is the current version before use.





CAUTION:

Do not isolate the chassis of item X01 from earth ground.

NOTE: With item P07 set at 0.0 V connect points A and B to module pins per table V. Adjust voltage for item X01 reading of 26 mA. Item M02 reading shall meet the limits of table V.

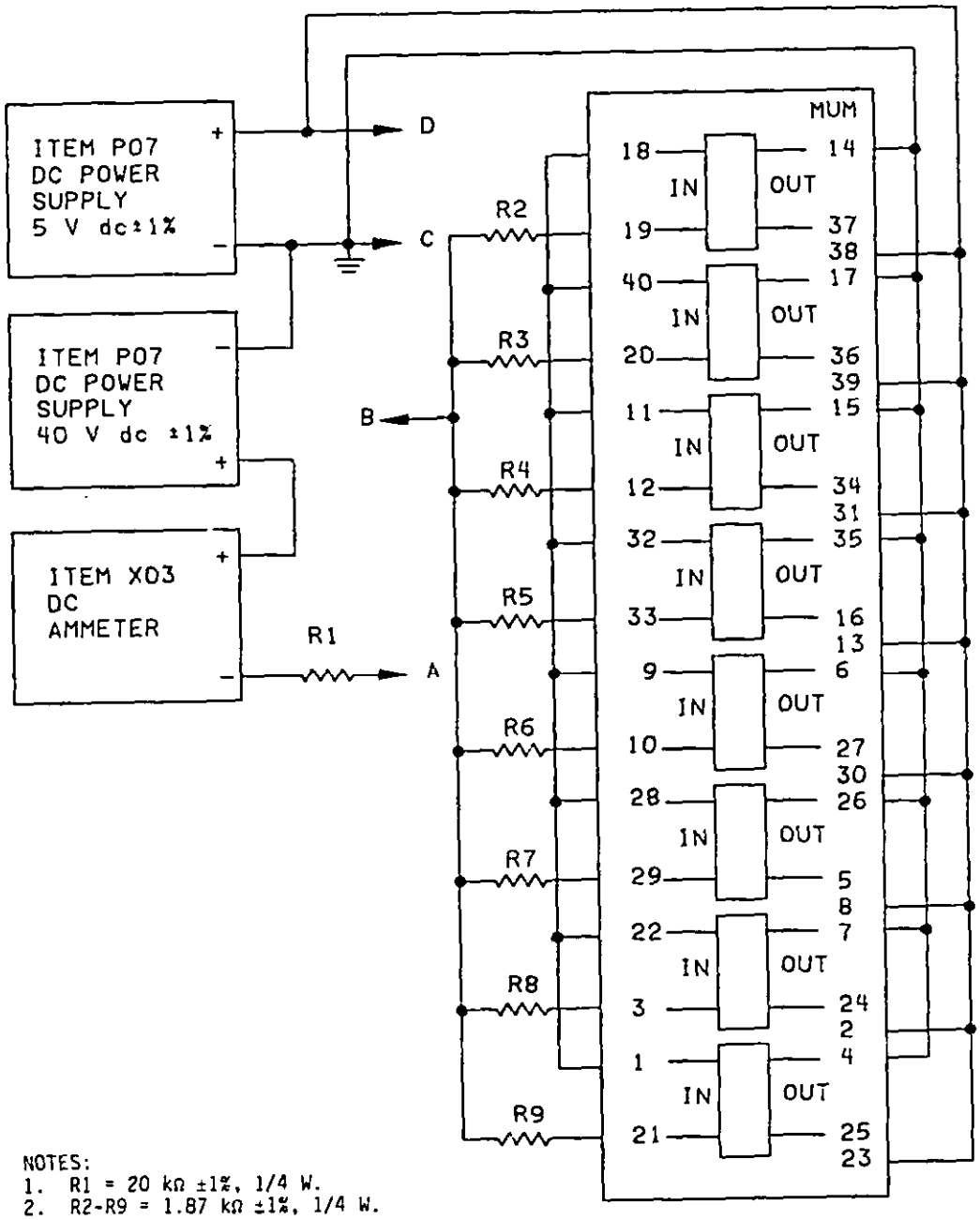
* FIGURE 9. LED forward voltage test circuit for table V.

Source: https://assist.dla.mil -- Downloaded: 202610-20T17:01Z
Check the source to verify that this is the current version before use.

* TABLE VI. Output leakage current and input threshold current test.

Test condition	Parameter	Connect A to pin		Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
E1	Output leakage current	37	C	0	40	µA	10
E2	"	36	"	"	"	"	"
E3	"	34	"	"	"	"	"
E4	"	16	"	"	"	"	"
E5	"	27	"	"	"	"	"
E6	"	5	"	"	"	"	"
E7	"	24	"	"	"	"	"
E8	"	25	"	"	"	"	"
E9	Input threshold current (If = 1 mA)	37	D	0.1	2.2	mA	"
E10	"	36	"	"	"	"	"
E11	"	34	"	"	"	"	"
E12	"	16	"	"	"	"	"
E13	"	27	"	"	"	"	"
E14	"	5	"	"	"	"	"
E15	"	24	"	"	"	"	"
E16	"	25	"	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-02-20T17:01Z
Check the source to verify that this is the current version before use.



NOTES:
1. R1 = 20 kΩ ±1%, 1/4 W.
2. R2-R9 = 1.87 kΩ ±1%, 1/4 W.

* FIGURE 10. Output leakage current and input threshold test circuit for table VI.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:01Z
Check the source to verify that this is the current version before use.

TABLE VII. LED reverse current test.

Test condition	Parameter	Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
F1	LED reverse current	18	0	150	uA	11
F2	"	40	"	"	"	"
F3	"	11	"	"	"	"
F4	"	32	"	"	"	"
F5	"	9	"	"	"	"
F6	"	28	"	"	"	"
F7	"	22	"	"	"	"
F8	"	1	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 20260-20T17:01Z
Check the source to verify that this is the current version before use.

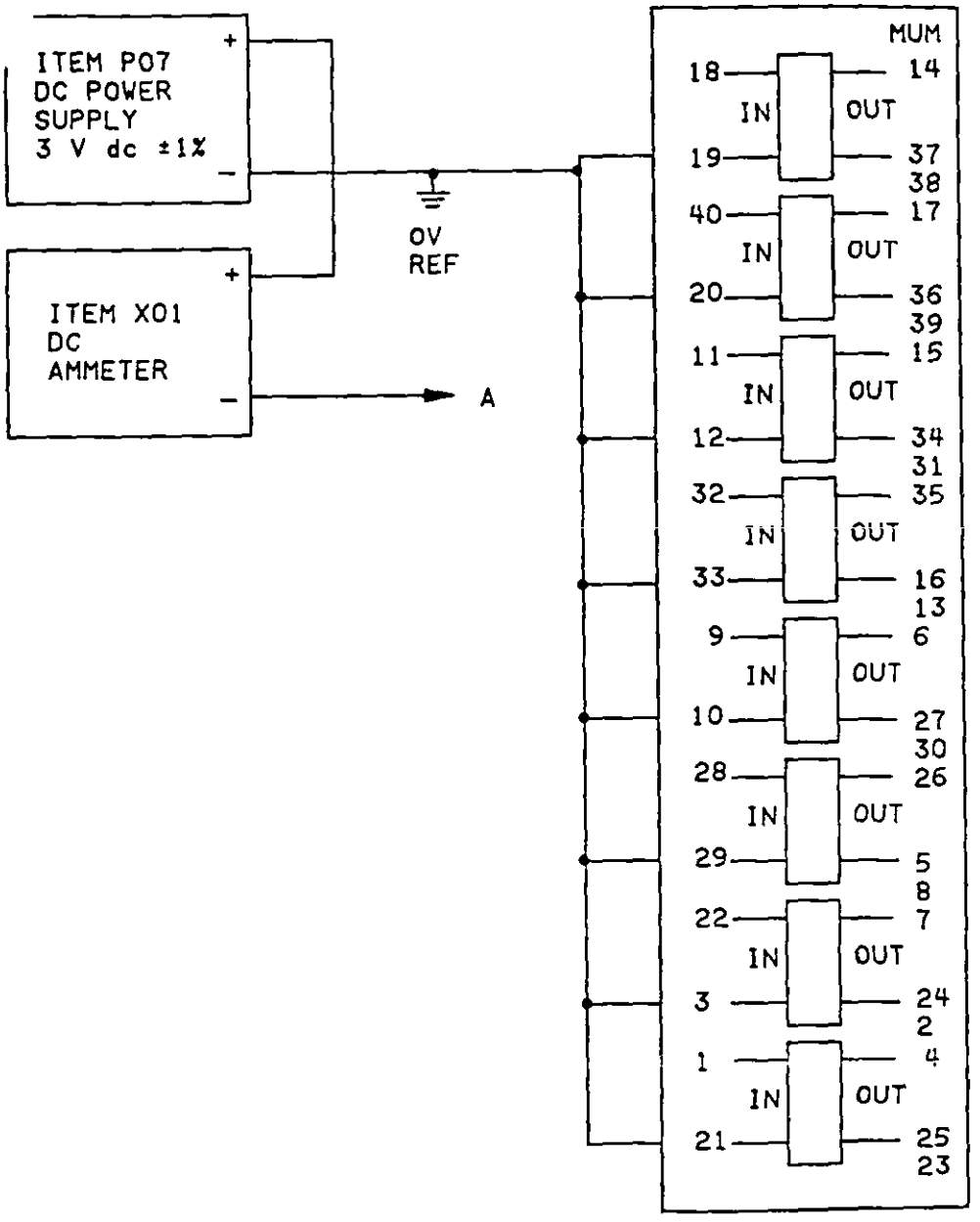


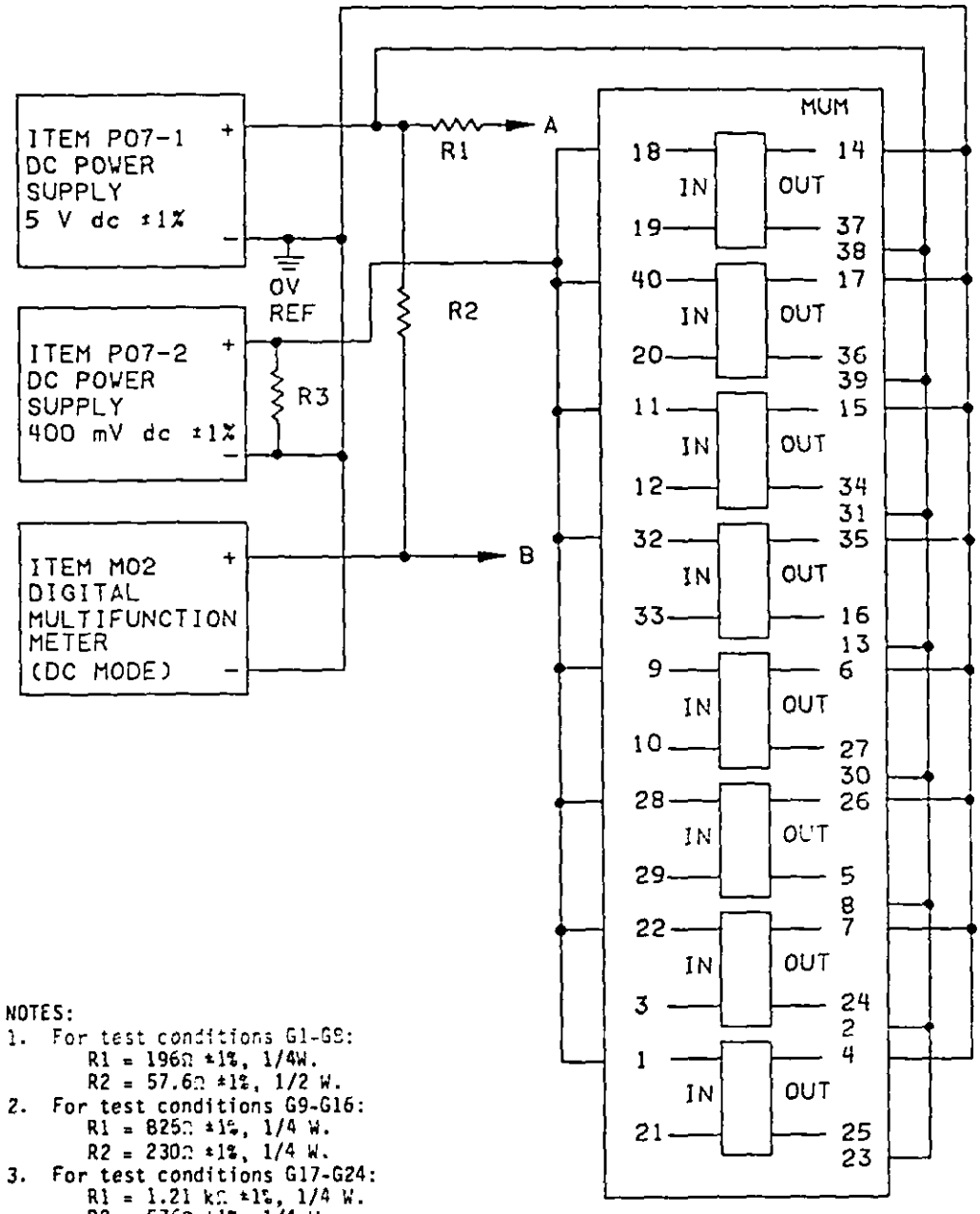
FIGURE 11. LED reverse current test circuit for table VII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:01Z
Check the source to verify that this is the current version before use.

* TABLE VIII. Output voltage (low-level) test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
G1	Output voltage, low level, I_{OL} = 80 mA, If = 16 mA	19	37	0	400	mV	12
G2	"	20	36	"	"	"	"
G3	"	12	34	"	"	"	"
G4	"	33	16	"	"	"	"
G5	"	10	27	"	"	"	"
G6	"	29	5	"	"	"	"
G7	"	3	24	"	"	"	"
G8	"	21	25	"	"	"	"
G9	Output voltage, low level, I_{OL} = 20 mA, If = 4 mA	19	37	"	"	"	"
G10	"	20	36	"	"	"	"
G11	"	12	34	"	"	"	"
G12	"	33	16	"	"	"	"
G13	"	10	27	"	"	"	"
G14	"	29	5	"	"	"	"
G15	"	3	24	"	"	"	"
G16	"	21	25	"	"	"	"
G17	Output voltage, low level, I_{OL} = 8 mA, If = 2.6 mA	19	37	"	"	"	"
G18	"	20	36	"	"	"	"
G19	"	12	34	"	"	"	"
G20	"	33	16	"	"	"	"
G21	"	10	27	"	"	"	"
G22	"	29	5	"	"	"	"
G23	"	3	24	"	"	"	"
G24	"	21	25	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-02-20T17:01Z
Check the source to verify that this is the current version before use.



NOTES:
1. For test conditions G1-G8:
   R1 = 196Ω ±1%, 1/4W.
   R2 = 57.6Ω ±1%, 1/2 W.
2. For test conditions G9-G16:
   R1 = 825Ω ±1%, 1/4 W.
   R2 = 230Ω ±1%, 1/4 W.
3. For test conditions G17-G24:
   R1 = 1.21 kΩ ±1%, 1/4 W.
   R2 = 576Ω ±1%, 1/4 W.
4. R3 = 51.1Ω ±10%, 1/4 W.

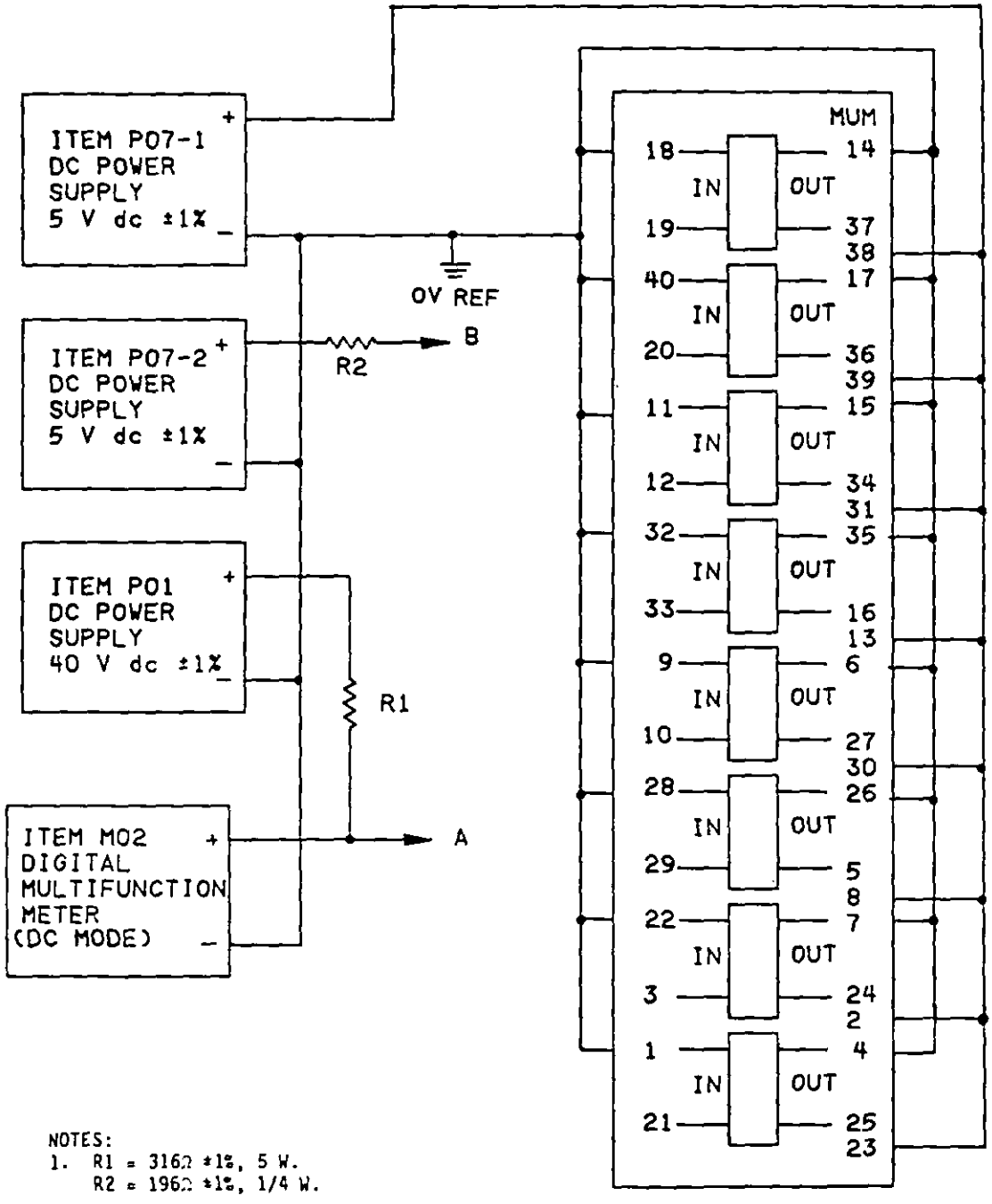
* FIGURE 12. Output voltage (low-level) test circuit for table VIII.

Source: https://assist.dla.mil -- Downloaded: 202610-20T17:01Z
Check the source to verify that this is the current version before use.

TABLE IX. Output saturation voltage test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
H1	Output saturation voltage (I_{OL} = 125 mA)	37	19	0	400	mV	13
H2	"	36	20	"	"	"	"
H3	"	34	12	"	"	"	"
H4	"	16	33	"	"	"	"
H5	"	27	10	"	"	"	"
H6	"	5	29	"	"	"	"
H7	"	24	3	"	"	"	"
H8	"	25	21	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-02-20T17:01Z
Check the source to verify that this is the current version before use.



NOTES:
1. R1 = 316Ω ±1%, 5 W.
   R2 = 196Ω ±1%, 1/4 W.

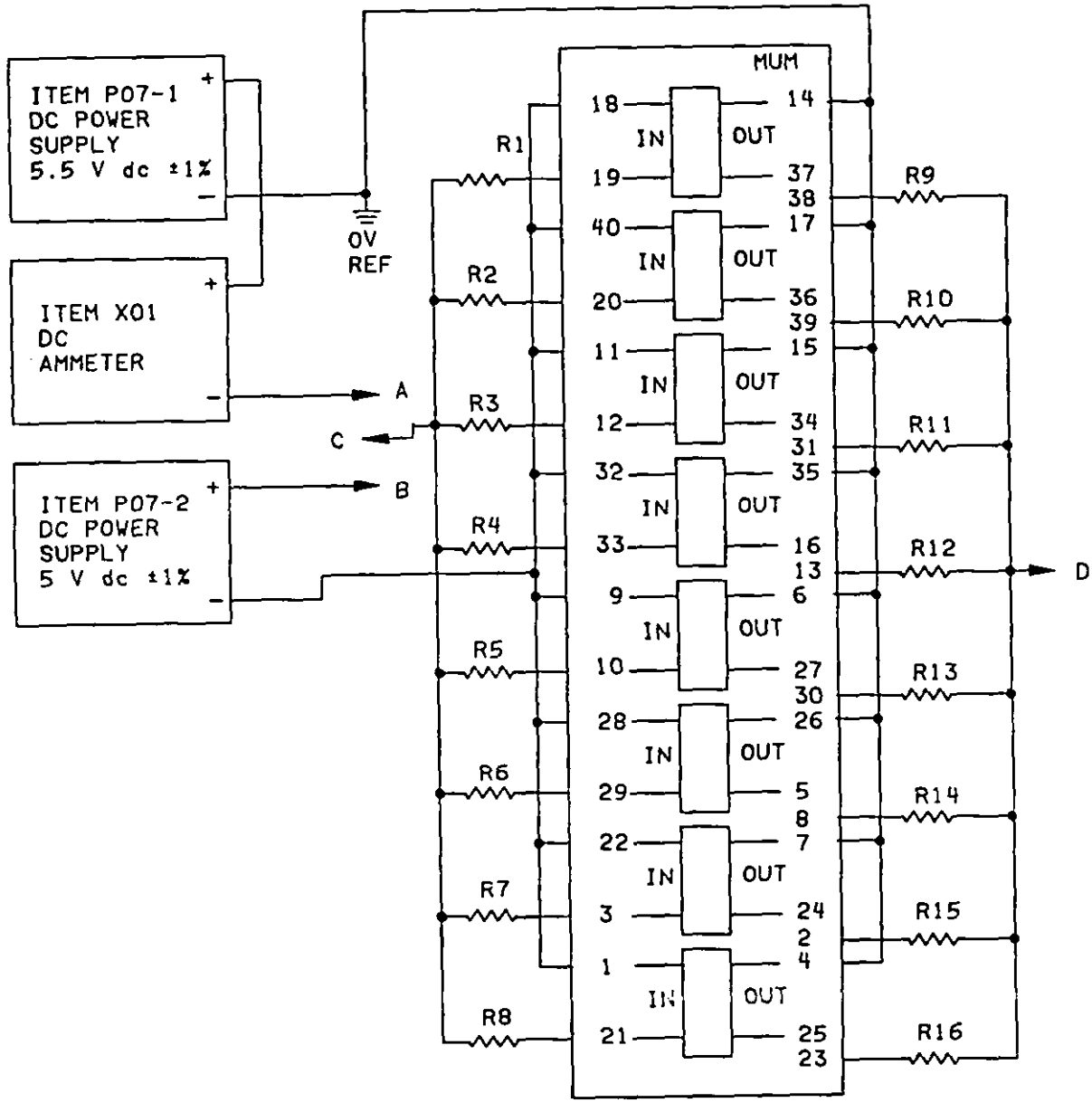
* FIGURE 13. Output saturation voltage test circuit for table IX.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:01Z
Check the source to verify that this is the current version before use.

TABLE X. Power supply current test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
J1	Power supply current, LED off	37, 36, 34, 16, 27, 5, 24, 25, D	Open	0	800	uA	14
J2	Power supply current, LED on	D	C	0	200	mA	14

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:01Z
Check the source to verify that this is the current version before use.



NOTES:
1. R1-R8 = 196Ω ±1%, 1/4 W.
2. R9-R16 = 220Ω ±1%, 1/4 W.

FIGURE 14. Power supply current test circuit for table X.

Source: https://assist.dla.mil -- Downloaded: 202640-20T17:01Z
Check the source to verify that this is the current version before use.

TABLE XI. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test.

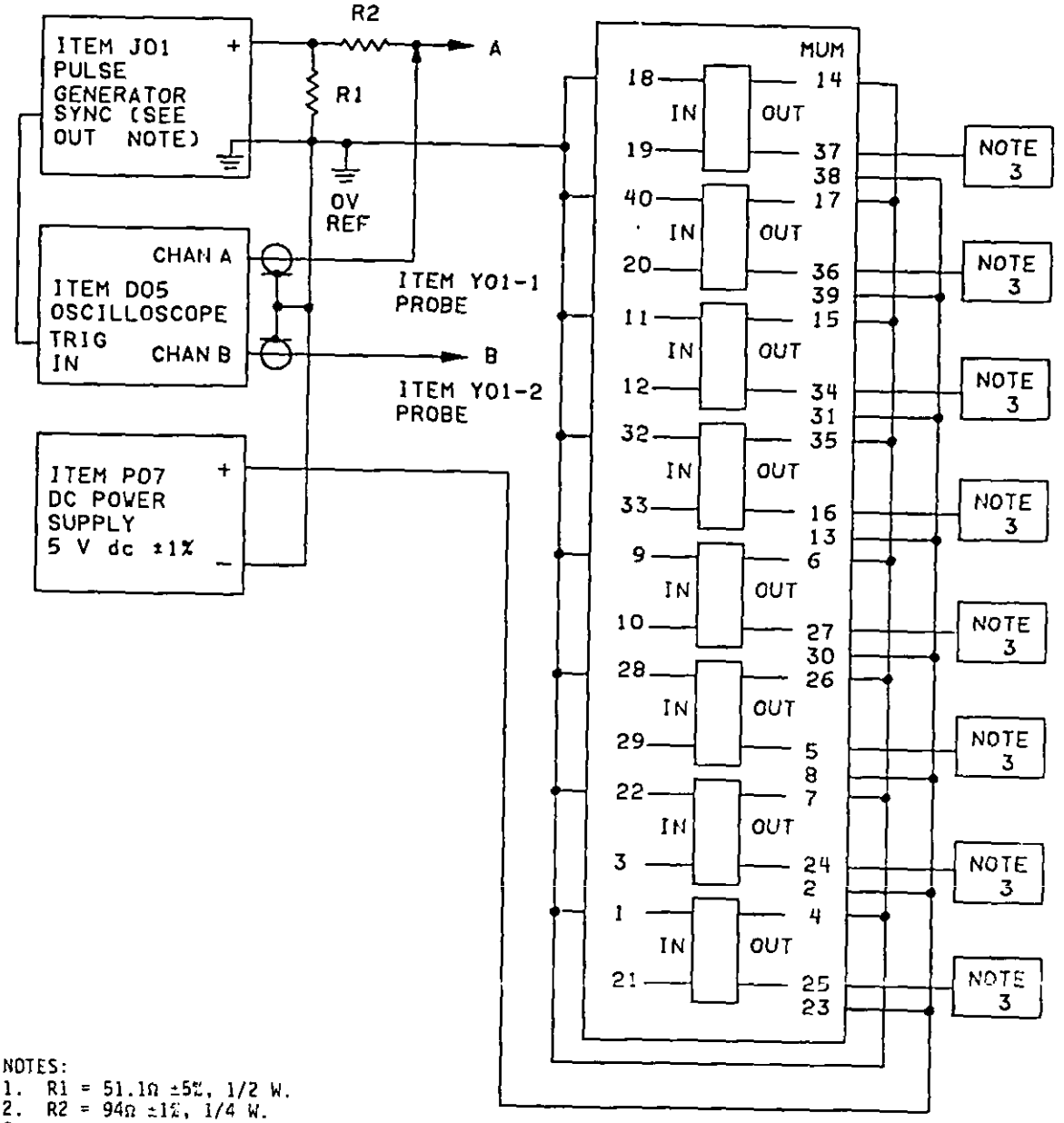
Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
K1	Transition time (t_{TLH})	19	37	1	12	µs	15, 16
K2	"	20	36	"	"	"	"
K3	"	12	34	"	"	"	"
K4	"	33	16	"	"	"	"
K5	"	10	27	"	"	"	"
K6	"	29	5	"	"	"	"
K7	"	3	24	"	"	"	"
K8	"	21	25	"	"	"	"
L1	Transition time (t_{THL})	19	37	0.060	1.400	"	"
L2	"	20	36	"	"	"	"
L3	"	12	34	"	"	"	"
L4	"	33	16	"	"	"	"
L5	"	10	27	"	"	"	"
L6	"	29	5	"	"	"	"
L7	"	3	24	"	"	"	"
L8	"	21	25	"	"	"	"
M1	Propagation delay time (t_{PLH})	19	37	1	14	"	"
M2	"	20	36	"	"	"	"
M3	"	12	34	"	"	"	"
M4	"	33	16	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-02-20T17:01Z
Check the source to verify that this is the current version before use.

TABLE XI. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
M5	Propagation delay time (t_{PLH})	10	27	1	14	µs	15, 16
M6	"	29	5	"	"	"	"
M7	"	3	24	"	"	"	"
M8	"	21	25	"	"	"	"
N1	Propagation delay time (t_{PHL})	19	37	0.1	2.5	"	"
N2	"	20	36	"	"	"	"
N3	"	12	34	"	"	"	"
N4	"	33	16	"	"	"	"
N5	"	10	27	"	"	"	"
N6	"	29	5	"	"	"	"
N7	"	3	24	"	"	"	"
N8	"	21	25	"	"	"	"

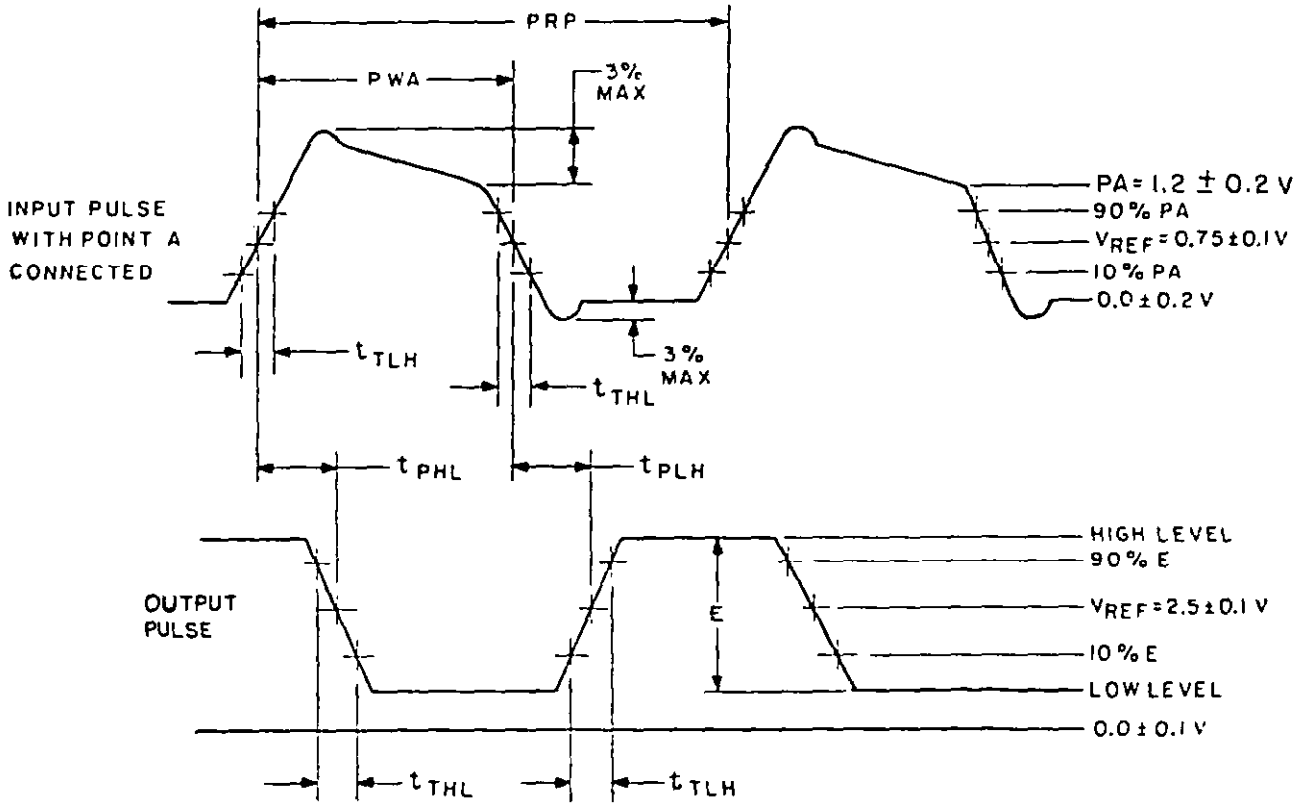
Source: https://assist.dla.mil -- Downloaded: 2026-02-20T17:01Z
Check the source to verify that this is the current version before use.



NOTES:
1. R1 = 51.1Ω ±5%, 1/2 W.
2. R2 = 94Ω ±1%, 1/4 W.
3. Load all outputs as per figure 17.
4. Set item J01 for 3.0 ±0.1 V at the pulse generator side of R2.

FIGURE 15. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit for table XI.

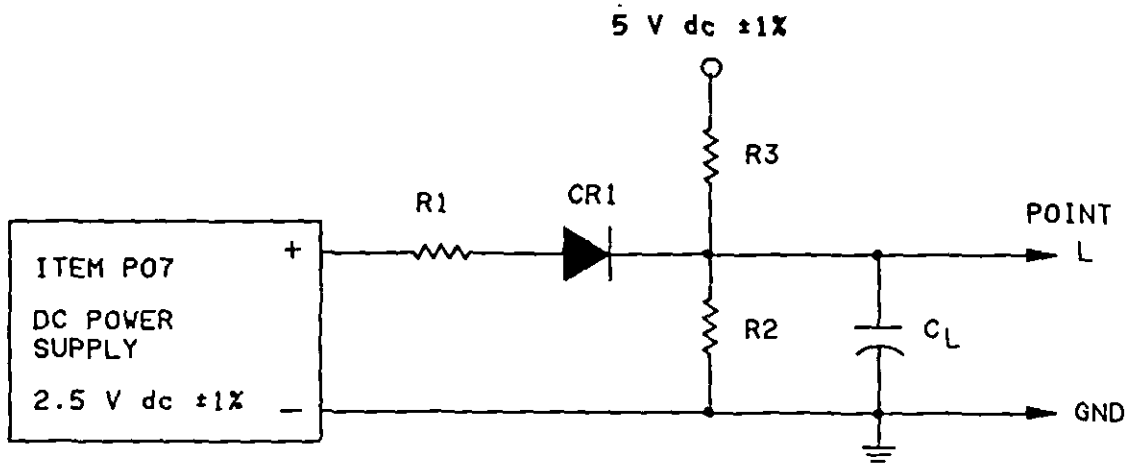
Source: https://assist.dla.mil -- Downloaded: 20260-20T17:01Z
Check the source to verify that this is the current version before use.



NOTES:
1. Input pulse parameters:
   PWA = 20 ±5 µs.
   PRP = 40 ±1 µs.
   t_{TLH} = 0.55 ±0.05 µs, t_{THL} = 0.8 ±0.1 µs.
2. All voltage levels are referenced to 0 volt.

FIGURE 16. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for table XI.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:01Z
Check the source to verify that this is the current version before use.



NOTES:
1. CR1 = 1N4150 or equivalent.
2. R1 = Select value for a current flow of 125 mA ±5% out of point L with point L held at 0.4 V dc ±1%.
3. R2 = 5.11 kΩ ±1%, 1/4 W.
4. C_L = 50 ±5 pF.
5. R3 = 2.2 kΩ ±1%, 1/4 W.
6. The maximum lead length from point L to the appropriate module pin shall be less than 1 inch (25.4 mm).

* FIGURE 17. MUM logic load circuit.

Source: https://assist.dla.mil -- Downloaded: 202610-20T17:01Z
Check the source to verify that this is the current version before use.

TABLE XII. Output crosstalk (low-level) test.

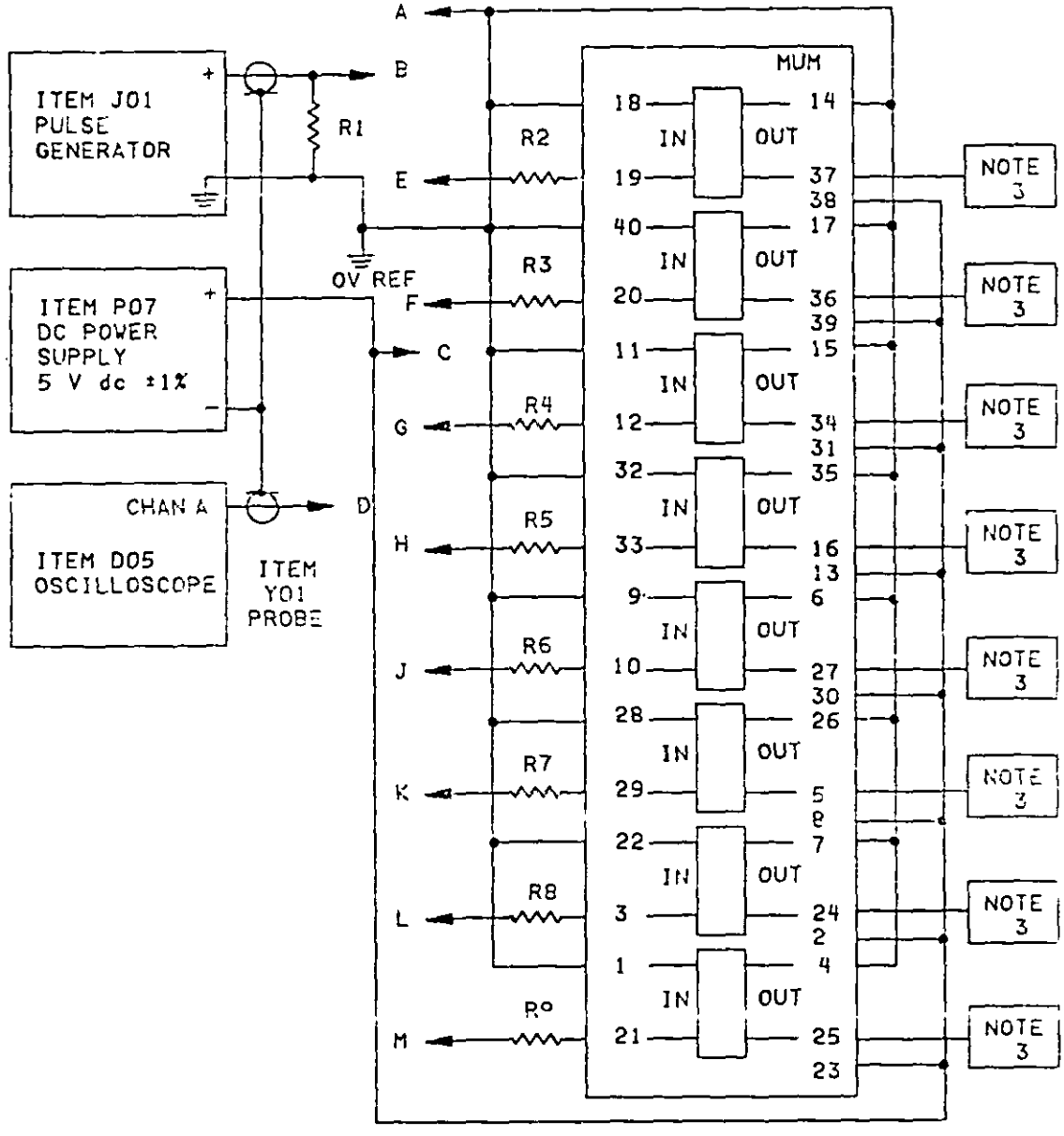
Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
P1	Out 1	Output crosstalk, low level	Open	F, G, H, J, K, L, M	E	37	0	400	mV	18, 19
P2	Out 2	"	"	E, G, H, J, K, L, M	F	36	"	"	"	"
P3	Out 3	"	"	E, F, H, J, K, L, M	G	34	"	"	"	"
P4	Out 4	"	"	E, F, G, J, K, L, M	H	16	"	"	"	"
P5	Out 5	"	"	E, F, G, H, K, L, M	J	27	"	"	"	"
P6	Out 6	"	"	E, F, G, H, J, L, M	K	5	"	"	"	"
P7	Out 7	"	"	E, F, G, H, J, K, M	L	24	"	"	"	"
P8	Out 8	"	"	E, F, G, H, J, K, L	M	25	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-02-20T17:01Z
Check the source to verify that this is the current version before use.

TABLE XIII. Output crosstalk (high-level) test.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
Q1	Out 1	Output crosstalk, high level	E	F, G, H, J, K, L, M	Open	37	0	400	mV	18, 19
Q2	Out 2	"	F	E, G, H, J, K, L, M	"	36	"	"	"	"
Q3	Out 3	"	G	E, F, H, J, K, L, M	"	34	"	"	"	"
Q4	Out 4	"	H	E, F, G, J, K, L, M	"	16	"	"	"	"
Q5	Out 5	"	J	E, F, G, H, K, L, M	"	27	"	"	"	"
Q6	Out 6	"	K	E, F, G, H, J, L, M	"	5	"	"	"	"
Q7	Out 7	"	L	E, F, G, H, J, K, M	"	24	"	"	"	"
Q8	Out 8	"	M	E, F, G, H, J, K, L	"	25	"	"	"	"

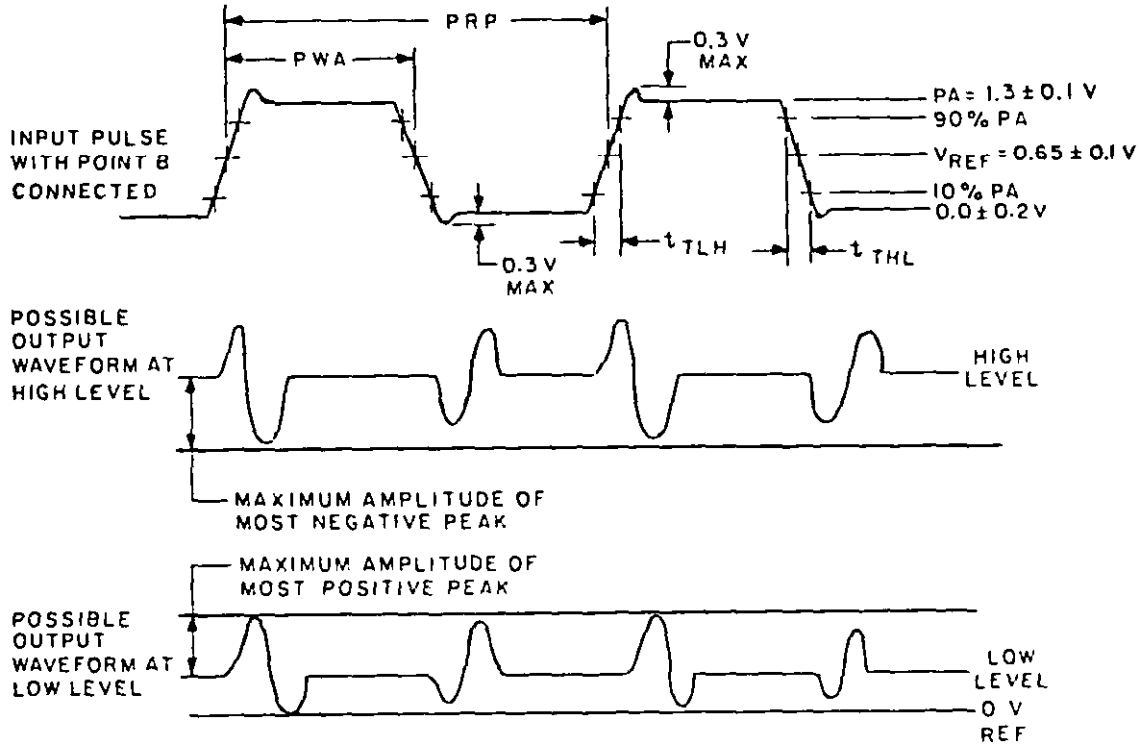
Source: https://assist.dla.mil -- Downloaded: 2026-02-20T17:01Z
Check the source to verify that this is the current version before use.



NOTES:
1. R1 = 51.1Ω ±5%, 1/2 W.
2. R2 - R9 = 133Ω ±1%, 1/4 W.
3. Load outputs wth logic load as shown on figure 17.

FIGURE 18. Output crosstalk (low and high level) test circuit for tables XII and XIII.

Source: https://assist.dla.mil -- Downloaded: 20260-20T17:01Z
Check the source to verify that this is the current version before use.



NOTES:
1. Input pulse parameters: For each test condition with the module connected to item J01, adjust item J01 for the required pulse amplitude. Pulse low level voltage and transition times. Verify the input pulse parameters at point B before making each measurement.
   $t_{TLH} = t_{THL} = 1.0 \pm 0.5$ µs.
   PWA = 20 ±5 µs.
   PRP = 40 ±1 µs.
2. Verify that the output voltage low and high level are $\leq$ 400 mV and $\geq$ 2.4 V respectively.

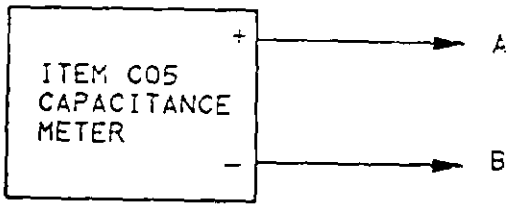
* FIGURE 19. Output crosstalk (low and high level) test circuit for tables XII and XIII.

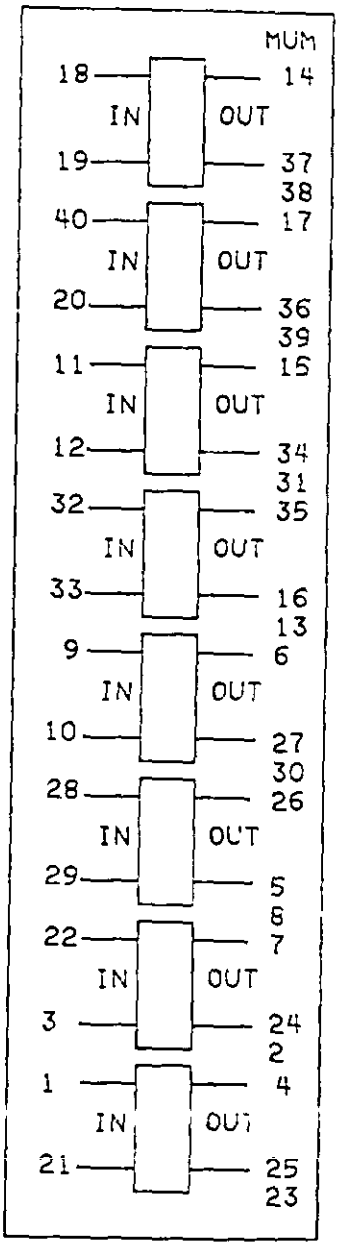
Source: https://assist.dla.mil -- Downloaded: 202610-20T17:01Z
Check the source to verify that this is the current version before use.

TABLE XIV. Input capacitance test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
R1	Input capacitance	21	1	1	200	pF	20
R2	"	3	22	"	"	"	"
R3	"	29	28	"	"	"	"
R4	"	10	9	"	"	"	"
R5	"	33	32	"	"	"	"
R6	"	12	11	"	"	"	"
R7	"	20	40	"	"	"	"
R8	"	19	18	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-02-20T17:01Z
Check the source to verify that this is the current version before use.





NOTES:
1. Null out the effects of the test leads prior to connecting the module.
2. Do not attach test connector or wires to any of the module pins other than the pins under test. Stray capacitance added to the other pins may affect the readings.

FIGURE 20. Input capacitance test circuit for table XIV.

Source: https://assist.dla.mil -- Downloaded: 20260-20T17:01Z
Check the source to verify that this is the current version before use.

6.6 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Preparing activity:
Navy - EC

Agent:
DLA - ES

(Project 5963-N699)

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:01Z
Check the source to verify that this is the current version before use.